# AI-Based Tropical Cyclone Intensification Prediction for the Bay of Bengal

A Multi-Modal Machine Learning Early Warning System

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## Abstract

Tropical cyclones in the Bay of Bengal cause catastrophic loss of life and infrastructure across Bangladesh, India, and Myanmar. Accurate prediction of cyclone intensification — whether an existing storm will significantly strengthen — remains one of the most challenging problems in operational meteorology. This paper presents a three-version AI-based early warning system that progressively incorporates richer data modalities: historical track features (V1), atmospheric reanalysis (V2), and satellite infrared image sequences (V3). Experiments on 375 Bay of Bengal storms (1970–2024) from IBTrACS and ERA5 reanalysis data (2018–2020) show that each additional data modality improves prediction accuracy, with ROC-AUC improving from 0.58 (V1) to 0.67 (V2) to 0.89 (V3) at the 48–72 hour forecast horizon. A real-time Streamlit dashboard integrates live weather from Open-Meteo API and active storm alerts from GDACS to enable operational use without any manual data entry.

## 1. Introduction

The Bay of Bengal (BoB) accounts for approximately 7% of global tropical cyclone activity but disproportionately high mortality. The funnel-shaped coastline of Bangladesh, shallow bathymetry, and dense coastal populations amplify storm surge impacts. Historical disasters include Cyclone Sidr (2007, ~3,500 deaths), Cyclone Fani (2019, ~89 deaths), and Cyclone Amphan (2020, ~128 deaths, $13 billion damage).

The central challenge addressed here is **intensification prediction**: given a cyclone's current state, will it significantly strengthen over the coming days? This work uses a threshold of ≥15 knots wind increase within a forecast window — applicable to broader intensification at longer lead times than the standard Rapid Intensification definition.

**Research Question:** Does integrating atmospheric reanalysis (ERA5) and satellite infrared image sequences with historical cyclone-track data improve prediction of tropical cyclone intensification in the Bay of Bengal at 48–72 hour and 120–168 hour lead times?

**H1:** Adding ERA5 atmospheric variables (V2) improves over track-only baseline (V1).
**H2:** Adding satellite infrared image sequences (V3) further improves over V2.

## 2. Datasets

### 2.1 IBTrACS v04r01 — Historical Track Data

The International Best Track Archive for Climate Stewardship (Knapp et al., 2010) provides 6-hourly cyclone observations for the North Indian Ocean from 1970 to 2024. After filtering to the BoB domain (0–25°N, 80–100°E), 375 storms were used for training V1, comprising 12,847 individual observations. Variables include position (lat/lon), wind speed (knots), pressure, and storm category.

### 2.2 ERA5 Reanalysis — Atmospheric Data

ERA5 hourly single-level reanalysis (Hersbach et al., 2020) was downloaded via the Copernicus Climate Data Store API for 2018–2020, covering the BoB at 0.25° resolution. Three NetCDF files per year (accumulated, instantaneous, wave) were merged using xarray. Each IBTrACS observation was matched to the nearest ERA5 grid point, yielding 520 matched observations across 21 storms.

| Variable | ERA5 Name | Unit | Role |
|---|---|---|---|
| Sea surface temperature | sst | °C | Primary fuel source |
| Mean sea-level pressure | msl | hPa | Ambient pressure |
| 10m U/V wind components | u10, v10 | m/s | Wind shear proxy |
| 2m temperature | t2m | °C | Atmospheric instability |
| 2m dewpoint | d2m | °C | Moisture content |
| Total precipitation | tp | m | Convective activity |

### 2.3 GridSat-B1 — Satellite Infrared Imagery

GridSat-B1 (Knapp et al., 2011) provides 3-hourly global infrared brightness temperature composites from 1980 to present, freely available via NOAA AWS S3. The infrared window channel (ch4, ~11 µm) captures cloud-top temperatures. A total of 445 images were downloaded across 9 major storms: MALA (2006), SIDR (2007), GIRI (2010), PHAILIN (2013), HUDHUD (2014), TITLI (2018), FANI (2019), AMPHAN (2020), MOCHA (2023). Each observation uses a 4-frame sequence (T-18h, T-12h, T-6h, T-0h) of 64×64 pixel IR patches centered on the storm.

### 2.4 Live Data for Operational Dashboard

**Open-Meteo API** (open-meteo.com): Free, no authentication. Provides current SST, surface pressure, 2m temperature, dewpoint, 10m wind speed/direction, and precipitation for any lat/lon in the Bay of Bengal. Used as ERA5 proxy for real-time V2 inference.
**GDACS RSS Feed** (gdacs.org): Active tropical cyclone alerts filtered to BoB region (0–25°N, 80–100°E). Provides storm name, position, wind speed, and alert level.

## 3. Methodology

### 3.1 Label Generation

For each observation i, the intensification label TARGET ∈ {0,1} is defined as:

```
TARGET = 1 if max(WIND in [t+h_start, t+h_end]) − WIND_i ≥ 15 kt
TARGET = 0 otherwise
TARGET = NaN if no future observations in window
```

| Label | Window | Meaning |
|---|---|---|
| Short | 48–72 hours | Will the cyclone intensify in 2–3 days? |
| Long | 120–168 hours | Will the cyclone intensify in 5–7 days? |

### 3.2 Train-Test Split

Storm-grouped splitting (GroupShuffleSplit, test_size=0.25) ensures all observations from a given storm are assigned exclusively to either train or test, preventing data leakage. The same storm cannot appear in both sets.

### 3.3 Version 1 — XGBoost Baseline

XGBoost classifier (Chen & Guestrin, 2016) trained on 9 track features with median imputation. Hyperparameters: n_estimators=300, max_depth=4, learning_rate=0.05. Training data: 375 BoB storms (1970–2024).

Features: LAT, LON, WIND, MONTH, HOUR, LAT_CHANGE, LON_CHANGE, WIND_CHANGE, TRACK_SPEED

### 3.4 Version 2 — XGBoost + ERA5

V1 extended with 7 ERA5 variables matched to each storm observation, giving 16 total features. Training data: 21 storms from 2018–2020 (520 observations with full ERA5 coverage). Same XGBoost pipeline. In the dashboard, ERA5 variables are replaced with real-time Open-Meteo data for live inference.

### 3.5 Version 3 — CNN-LSTM Fusion

A two-branch PyTorch neural network processes satellite image sequences and tabular track features simultaneously:

```
Branch A — Images: (4, 64, 64) IR sequence
→ Conv2D(1→16) + BN + ReLU + MaxPool(4)
→ Conv2D(16→32) + BN + ReLU + MaxPool(4)
→ Flatten → Linear(512→64) → LSTM(64→32)

Branch B — Track features: Linear(9→32→16)

Fusion: Concat[32+16] → Linear(48→32→1) → BCEWithLogitsLoss
```

Training: Adam optimizer (lr=5×10■■), CosineAnnealingLR, early stopping (patience=15), class-weighted loss. 9 storms, 298 training samples.

## 4. Results

### 4.1 Main Comparison

| Version | Data | ROC-AUC 48–72h | ROC-AUC 120–168h | PR-AUC 48–72h | Brier Score |
|---|---|---|---|---|---|
| V1 XGBoost | Track only | 0.5848 | 0.1000 | 0.6242 | 0.299 |
| V2 XGBoost | Track + ERA5 | 0.6696 | 0.8037 | 0.7703 | 0.269 |
| V3 CNN-LSTM | Track + Satellite | 0.8862 | — | 0.9487 | 0.143 |

Both H1 and H2 are confirmed. ERA5 atmospheric variables dramatically improve the 5–7 day prediction (V1: 0.10 → V2: 0.80), confirming that large-scale thermodynamic conditions dominate at longer lead times. Satellite image sequences provide the largest single improvement at the 2–3 day horizon (V2: 0.67 → V3: 0.89).

### 4.2 V3 Classification Report (48–72h, n=83)

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Non-intensifying (0) | 0.65 | 0.81 | 0.72 | 27 |
| Intensifying (1) | 0.90 | 0.79 | 0.84 | 56 |
| Weighted Average | 0.82 | 0.80 | 0.80 | 83 |

Brier Score: 0.143 (well-calibrated; perfect = 0.0, random = 0.25)

## 5. Operational Dashboard

A Streamlit web dashboard enables real-time intensification risk assessment:

| Feature | Description |
|---|---|
| GDACS Integration | Detects active BoB cyclones in real-time; auto-fills position |
| Open-Meteo Integration | Fetches live SST, pressure, wind, temp for any position |
| V1 → V2 Auto-upgrade | Switches to V2 model automatically when live weather is loaded |
| Interactive Map | Folium map with Esri Ocean basemap, direction arrow, storm popup |
| Risk Display | LOW / MODERATE / HIGH badges with probability percentage |
| Historical Tracks | Last 10 IBTrACS storms overlaid as reference |

## 6. Limitations

**ERA5 is reanalysis, not forecast:** V2 uses reconstructed past conditions. For true operational use, replace ERA5 with GFS or ECMWF forecast fields.

**Small ERA5 training set:** Only 21 storms (2018–2020) used for V2 due to manual download requirement. Extending to 2010–2020 would improve generalization.

**V3 long-range gap:** 120–168h V3 evaluation unavailable — test split contained only intensifying storms (single class) due to 9-storm dataset size.

**Intensification only:** The system predicts whether a cyclone strengthens, not track, landfall location, rainfall, or flood extent.

**Open-Meteo vs ERA5 resolution:** Live weather uses Open-Meteo as ERA5 proxy. Spatial resolution differences (~0.1° vs 0.25°) may introduce slight bias.

## 7. Conclusion

This work demonstrates that multi-modal data integration significantly improves tropical cyclone intensification prediction in the Bay of Bengal. The progressive improvement across V1 → V2 → V3 confirms both research hypotheses. ROC-AUC improved from 0.58 to 0.89 at the 48–72 hour horizon, and from 0.10 to 0.80 at the 120–168 hour horizon by adding ERA5 atmospheric data.

The operational dashboard bridges research prototype and real-world use by automatically fetching live atmospheric data and active storm positions, enabling zero-manual-input risk assessment during an active cyclone event.

**Future work:** (1) Expand ERA5 coverage to 2010–2020 for V2; (2) Integrate INSAT-3D from MOSDAC for higher-resolution Indian Ocean satellite data; (3) Replace ERA5 with operational NWP forecast fields for true real-time V2 inference; (4) Extend to predict landfall intensity and storm surge.

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